Code No: 115DT **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, January/February - 2023
COMPUTER NETWORKS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

JNTUH Used papers 2023

PART - A

(25 Marks)

1.a)	What are the uses of Computer Network?	[2]
b)	What is piggy backing technique? When and why is it used?	[3]
c)	What is bit stuffing? Give an example.	[2]
d)	What is the use of gateway and router?	[3]
e)	What is optimality principle?	[2]
f)	Write a brief note on store and forward packet switching.	[3]
g)	What is tunneling?	[2]
h)	Give a short note on IP addresses.	[3]
i)	What is the function of transport layer?	[2]
j)	What is the use of HTTP and FTP protocols?	[3]

PART - B

(50 Marks)

2.	Explain various layers and their functions of TCP/IP reference model.	[10]
	OR	
3.	Explain the stop-and-wait protocol and selective repeat protocol.	[10]
4.	Explain the CSMA/CD protocol in detail.	[10]
	OR	
5.	Describe the need and use of various network devices.	[10]
6.	Explain the working principle of the Hierarchical routing protocol.	[10]
	OR	
7.	Explain the token bucket algorithm. Write down its advantages and disadvantages.	[10]
8.	Discuss IMCP and DHCP protocols in detail.	[10]
	OR	
9.	Explain the elements of transport protocol.	[10]
10.a)	Explain the TCP service model.	
b)	Draw the TCP header and explain various fields.	[5+5]
	OR	
11.	Explain the TELNET and DNS in detail.	[10]

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